DATADOSE
PROJECT DESCRIPTION

AI-powered polypharmacy safety platform

Project Overview

DataDose is an enterprise-grade, AI-powered Clinical Decision Intelligence Platform that unifies graph database technology, real-time data streaming, and generative AI to deliver polypharmacy safety directly at the point of care. The platform operates across three deeply integrated layers: a live clinical application (Next.js 15 + FastAPI + Neo4j AuraDB) serving five role-based dashboards, a complete data engineering pipeline (pandas → OpenFDA/Groq verification → Aiven Kafka → Databricks PySpark → Snowflake → Power BI), and an Apache Airflow orchestration layer with its own PostgreSQL metadata store that schedules, triggers, and monitors every pipeline stage end to end. All application services are containerized with Docker Compose, and every layer of the platform — from the live clinical assistant to the analytics warehouse — is unified around the same underlying pharmaceutical knowledge graph, ensuring clinical intelligence and population-level analytics stay conceptually consistent rather than operating as disconnected systems.

Problem Statement

Polypharmacy — patients taking five or more concurrent medications — affects over 40% of elderly patients and is a leading cause of preventable adverse drug events. Clinicians often lack real-time, context-aware tools to check interactions across a patient's complete medication list during a clinical encounter, forcing decisions to be made without a full, systematic view of drug-drug, drug-disease, and drug-symptom risk. At the same time, healthcare organizations need to aggregate and analyze prescription patterns across entire patient populations, but raw EHR and pharmacy data is typically fragmented, unvalidated, and siloed away from analytics infrastructure — making it difficult to turn individual clinical events into governed, trustworthy, population-level insight.

Solution

DataDose embeds a Neo4j AuraDB drug-interaction knowledge graph — spanning over 3,656 Drug, Disease, and Symptom nodes connected by INTERACTS_WITH, TREATS, and CAUSES_REACTION relationships — directly into the clinical workflow. Sub-second polypharmacy checks, AI-powered safe alternatives, multilingual symptom tracing, prescription OCR, and a hybrid GraphRAG assistant are surfaced through five role-specific dashboards for Patients, Physicians, Pharmacists, Admins, and Super Admins. Behind the scenes, a five-stage validation pipeline (pandas cleaning → OpenFDA cross-reference → Groq LLM verification → Aiven Kafka streaming → Databricks PySpark enrichment) produces clean, risk-scored prescription events that land in a governed four-layer Snowflake star schema, feeding executive Power BI dashboards. An Apache Airflow control plane ties the whole pipeline together, verifying infrastructure health, triggering the Databricks job, promoting staging data into the warehouse, running data quality checks, and sending Slack notifications — so every stage is both triggerable and monitorable rather than run as isolated manual scripts.

Key Features

- **Real-Time DDI Scanning:** N-degree polypharmacy scanner that checks every pairwise INTERACTS_WITH relationship among a patient's complete drug list via Neo4j Cypher traversal, returning results sorted by severity (Major, Moderate, Minor) in milliseconds — giving clinicians a full-regimen safety check rather than a two-drug lookup.
- **GraphRAG Clinical Assistant:** Hybrid chatbot that performs entity extraction on a clinician's natural-language question, retrieves deterministic relationships from the Neo4j graph, and then synthesizes a grounded response with LLaMA-3 — supporting multi-turn, context-aware clinical conversations while minimizing hallucination.
- **Smart Safe Alternatives:** Multi-constraint Cypher traversal that, given a drug to replace, a disease to treat, and a symptom to avoid, surfaces safe substitute medications from the graph — falling back to Groq LLM synthesis whenever the graph search returns zero results.
- **Prescription OCR Scanner:** A vision-language model (Groq's LLaMA 3.2 11B Vision) extracts drug names directly from uploaded handwritten or printed prescription images, feeding the results straight into the DDI scanner for an immediate safety check.
- **Multilingual Symptom Tracer:** A built-in English/Arabic synonym dictionary expands a reported adverse symptom across clinical and colloquial terms, then traces it back through the patient's current medication list to the most likely causative drug.
- **Visual Prescription Map:** An interactive React Flow canvas renders a patient's full drug list as an explorable knowledge graph, visualizing DDI edges, treated diseases, and caused symptoms in a single view clinicians can navigate.
- **Data Pipeline & Analytics:** A five-stage pipeline — pandas cleaning, OpenFDA/Groq verification, Aiven Kafka streaming, Databricks PySpark enrichment, and Snowflake warehousing — turns raw pharmacy records into risk-scored, analytics-ready data feeding a 4-page Power BI reporting suite (Home, Overview, Clinical Insights, Risk Analysis).

System Architecture

The DataDose platform is built across three integrated layers — live application, data pipeline, and orchestration:

1. **Live Clinical Application:** A Next.js 15 frontend and FastAPI backend, each containerized and orchestrated with Docker Compose, deliver real-time DDI scanning, an AI clinical assistant, and

five role-based dashboards (Patient, Physician, Pharmacist, Admin, Super Admin) over a REST API.
2. **Graph Database (Neo4j AuraDB):** Stores 3,656+ Drug, Disease, and Symptom nodes connected by INTERACTS_WITH, TREATS, and CAUSES_REACTION relationships, powering sub-second interaction queries via Cypher traversal for both the live application and the streaming enrichment job.
3. **Data Processing (Databricks PySpark):** Structured Streaming consumes Kafka prescription events on a 10-second micro-batch trigger, enriches each one against the Neo4j interaction graph, applies drug- and patient-level risk scoring, and writes to both Snowflake and a Kafka output topic — with malformed and failed records routed to a Delta dead-letter table instead of being silently dropped.
4. **Data Warehousing (Snowflake):** A governed four-layer star schema (staging → dimensions → facts → analytics) with SCD Type 2 history on DIM_DRUG and idempotent MERGE-based promotion, feeding the executive Power BI reporting suite.
5. **Orchestration (Airflow):** A local Airflow instance with its own PostgreSQL metadata DB runs a scheduled DAG (every 2 minutes, max_active_runs=1) across four task groups — infrastructure validation, Databricks processing, warehouse promotion, and data quality validation — and sends Slack notifications on both success and failure.
6. **Clinical Intelligence API:** 9 FastAPI REST endpoints covering DDI scanning, safe alternatives, symptom tracing, visual mapping, prescription OCR, and GraphRAG chat, backed by an async Neo4j driver and Groq LLM integration.
7. **Containerized Deployment:** Docker and Docker Compose orchestrate the frontend and backend services locally, alongside a separate Airflow container stack (webserver, scheduler, and Postgres) that manages the data pipeline independently.

Technology Stack

- **Graph Database:** Neo4j AuraDB
- **Streaming Platform:** Apache Kafka (Aiven-managed)
- **Big Data Processing:** Databricks (PySpark)
- **Data Warehouse:** Snowflake
- **Workflow Orchestration:** Apache Airflow
- **Programming Languages:** Python, TypeScript
- **AI / LLM:** Groq (LLaMA-3, LLaMA 3.2 11B Vision)
- **Containerization:** Docker, Docker Compose
- **API Framework:** FastAPI (Python) + Next.js 15 (TypeScript)

Project Metrics

DataDose delivers measurable impact through the following:

- **Real-Time DDI Coverage:** Sub-second pairwise interaction scanning across a patient's entire medication list, sorted automatically by clinical severity.

- **Knowledge Graph Scale:** 3,656+ nodes spanning Drug, Disease, and Symptom entities, linked by INTERACTS_WITH, TREATS, and CAUSES_REACTION relationships in Neo4j AuraDB.
- **API Coverage:** 9 REST endpoints spanning scanning, alternatives, symptom tracing, visualization, OCR, and GraphRAG chat, all served from a single FastAPI backend.
- **Streaming Cadence:** Databricks Structured Streaming processes Kafka events on a 10-second micro-batch trigger, carrying each record from the DataDose.in topic through Neo4j enrichment and risk scoring to Snowflake staging, with independent dead-letter handling for four distinct failure types.
- **Role-Based Access:** Five dedicated dashboards for Patient, Physician, Pharmacist, Admin, and Super Admin roles, each surfacing role-relevant clinical or administrative data.
- **Data Pipeline Throughput:** Configurable Kafka producer rate (messages/second, via --rate and --max flags) feeding a five-stage ETL from raw CSV through Kafka and Databricks PySpark to a governed, idempotent Snowflake warehouse, orchestrated on a 2-minute Airflow schedule.

Conclusion

The DataDose Clinical Decision Intelligence Platform demonstrates how graph databases, real-time streaming, workflow orchestration, and generative AI can combine into a single, cohesive system — from a live clinical application delivering sub-second polypharmacy safety checks, down to a production-quality data engineering pipeline that cleans, verifies, streams, enriches, and warehouses pharmaceutical data at scale. Every layer, from the FastAPI clinical endpoints to the Airflow-orchestrated Snowflake warehouse, is built with the same engineering discipline: encrypted credential management, idempotent and retry-safe operations, modular and independently testable components, and comprehensive observability through logging, dead-letter handling, and Slack alerting. Built with Next.js, FastAPI, Neo4j AuraDB, Kafka, Databricks, Snowflake, Power BI, Docker, and Apache Airflow, DataDose is designed for demonstration, education, and enterprise-architecture presentation purposes — illustrating how a real clinical safety problem can be addressed with a fully integrated, production-oriented technology stack.